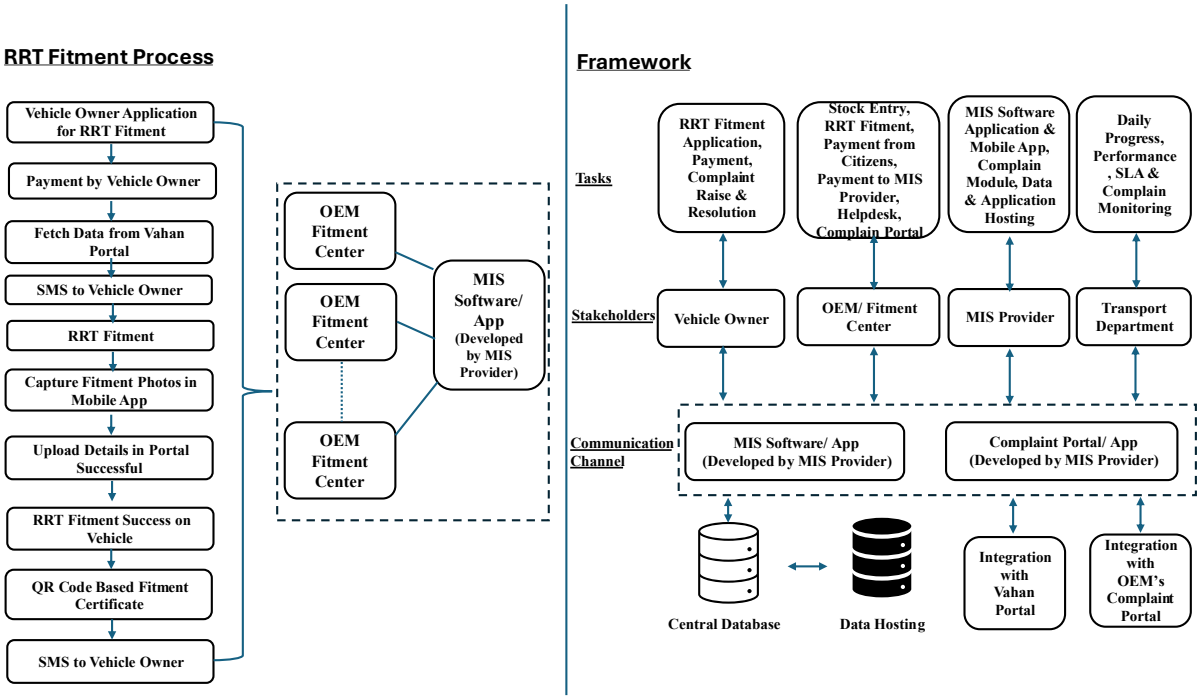
2 Objective of the Project

(i) To implement a transparent and efficient system for managing reflective tapes affixed to vehicles.
(ii) To develop a centralized digital platform for real-time data management, monitoring, certification and inventory management.
(iii) To generate secure QR code-based stickers for each vehicle reflecting vehicle and certification data.
(iv) To integrate with the **VAHAN 4.0** portal for syncing vehicle fitness and reflector data.
(v) To onboard all stakeholders including tape manufacturers, dealers, fitters, and transport officials onto one platform.
(vi) To provide an online complain redressal system for efficient grievance redressal.

3 Scope of Work

3.1 Indicative Process Flow



3.2 General Requirements of the Project

Sr.	Minimum Functional Requirement
Development & Deployment	
1.	The selected PSU will be responsible for end-to-end development and deployment of MIS console, QR-Code generation, Mobile & Web based application on the State Data Centre, or any reputed Data Centre duly empanelled with MEITY, Gov. of India or approved by the MPTD.
2.	All necessary Software, Licenses, SMS Cost etc. necessary for smooth development and subsequent operation of entire platform will be provided by the selected PSU.

Sr.	Minimum Functional Requirement
3.	Selected PSU shall also provide necessary training to officials of MPTD in all district offices and will ensure smooth timely rollout of updates in future.
4.	The PSU will ensure strict compliance with all applicable IT and data security laws and transport rules and regulations.
5.	The Selected PSU shall implement any change requests or updates issued by MPTD, and no additional cost shall be payable for such developments.
Login and Access Control	
6.	The MIS/ Software Application console should have separate role-based user onboardings/ logins for MPTD (Admin) and for empaneled Manufactures and dealers of reflective products with customization as decided by the MPTD.
7.	The application should have proper features for authentication and access control and should maintain detailed logs of these activities.
8.	The application should be available both in Web & Mobile based forms. The Application should be Secure, scalable and user-friendly.
QR Code Generation	
9.	The QR code affixed to the **Retro Reflective Tape Fitment Certificate must have** security features to guarantee its authenticity and trackability in case of misuse: i) **State code**: The state code embedded in the QR code ensures that it is generated for and used in the same state. This helps in regional tracking and verification. ii) **Manufacturer code:** The Manufacturer code within the QR code verifies that it is affixed to a reflector of the same Manufacturer for which is empaneled with the MPTD. iii) **Fitment Center Code** iv) **Month & Year of QR Code Generation and next inspection**: This element allows tracking of when the QR code was generated and when the next inspection needs to be conducted. This chronological information aids in monitoring and accountability, helping to detect any irregularities. v) **Provision for Verification of Genuine QR Code**. vi) **Vehicle Registration Details** (Reg. No., Name of Owner, Contact No, Validity etc.) as per VAHAN Portal.
Inventory Management & certificate Generation	
10.	The application should have a robust inventory management system where the manufacturer is able to allocate the supply of reflective products to dealers and monitors its consumption/sales in the State.
11.	The MIS Provider (PSU) will receive an API from the OEM containing complete details of each Retro-Reflective Tape QR code. This information will be automatically integrated into the PSU's software application through the API, ensuring accurate data transfer and eliminating discrepancies during verification at the Fitment Center.
12.	The system should ensure generation of QR code-based fitment certificate and use AI to assess the quantity of reflective products used and debit the same from stock of dealer on every fitment.
13.	The application should facilitate tracking of entire lifecycle of tape from manufacture to dealer till fitment.

Sr.	Minimum Functional Requirement
14.	The software application shall maintain a comprehensive repository and complete historical records of all vehicles on which RRT Fitment has been carried out. The system must be capable of securely storing and retrieving vehicle information details, including photographs captured after RRT Fitment process, date and time of fitment, vehicle registration number, vehicle category/type, contact number, address, and any other relevant data. This historical record will support verification, validation, audit, and compliance requirements as mandated by the department.
15.	• The MIS Provider shall develop a mechanism through which the vehicle owner can pay the required charges, RRT Charges, Service Charges, and QR Code Charges, directly to the OEM via various online (Internet Banking, Debit/ Credit etc.), offline payment methods. Additionally, the MIS Provider must create a digital wallet system within the software application, which the OEM will be responsible for maintaining and periodically recharging. The Wallet recharge facility shall be possible via online (Internet Banking, Debit/ Credit etc.) & offline payment methods. • For each QR-based Fitment Certificate issued to a vehicle owner, the applicable QR Code cost (as quoted during the bidding process) will be deducted from the OEM's wallet. The deducted amount will then be automatically transferred to the MIS Provider's account daily. • If a QR code becomes unreadable, contains no information, or displays incorrect information due to a software glitch, the application must regenerate a new QR code at no additional cost. • However, if the incorrect QR code is generated because the OEM provided wrong data, then the cost of regenerating the QR code shall be borne by the OEM, and the QR regeneration charges will also be credited to the MIS Provider.
16.	Once the Fitment Certificate is generated in the software application, an SMS notification with the Link to download the Fitment Certificate must be sent to the vehicle owner.
17.	The MIS portal shall provide an online facility via which the Vehicle owner can download the previously generated Fitment certificate free of cost, until certificate's validity.
Fitment Application at Fitness Center	
18.	The software application shall provide a facility for capturing new fitment application details for vehicles arriving at the designated fitment centers for RRT Fitment. The system must record all required information and documents submitted by the vehicle owner. After verification of the details, the authorized/approved RRT shall be affixed to the vehicle by the fitment center. Upon completion of the fitment, the system shall generate a QR-based Fitment Certificate, which will be issued to the vehicle owner for use in the subsequent vehicle fitness process.
Fitment Application from Remote	
19.	The software application should provide an online facility for vehicle owners to submit their RRT Fitment Application remotely. Through the system, the vehicle owner should be able to select a preferred fitment center and choose a convenient time slot. Based on the selected time slot, the vehicle owner can visit the chosen fitment center for the installation of the retro-reflective tape on the vehicle. After the fitment is completed, the fitment center should issue a Fitment Certificate, which the vehicle owner can use for the subsequent vehicle fitness process.

Sr.	Minimum Functional Requirement
Reports	
20.	The application should feature a highly customizable real time dashboard and report generation module, which the department or the manufactures or every dealer shall use to analyze the process of the installation & Stock of reflective products.
21.	The PSU shall obtain the prescribed format from the Authority and generate the report accordingly. The generated reports should be easily exportable in all commonly used formats.
Integration with VAHAN Portal	
22.	The system should integrate with the VAHAN Portal API to automatically capture vehicle details as required.
23.	The system should provide secured real time APIs to the VAHAN system, so that the fitment status is uploaded there and can be made use of by Automated Testing Stations (ATS) and RTO offices at the time of vehicle fitness.
24.	At the time of generation of Fitment certificate, the application should insist on OTP to be sent on registered Mobile Number of vehicle owner.
Technical support and Helpdesk	
25.	Selected PSU will ensure 24x7 technical support and maintain a central helpdesk in premises provided by MPTD along with all necessary equipment, furniture, peripherals and adequate and trained Manpower.
Complaint portal	
26.	Selected PSU shall develop the module to enable the Vehicle owner/Manufacturer /Dealers to lodge complaints relating to the services provided.
27.	The Selected PSU will also get the API from the OEMs to get all the Complain status logged & resolved at OEMs End (Complaints from OEMs Complain Portal)
28.	The link for such module shall be provided to the transport department for display on its website
29.	The MIS reports regarding the status of pending complaints shall be given to the MPTD regularly as per the provisions of SLA.
Manpower	
30.	For Efficient operation of the system Selected PSU will provide Software & Hardware as per specification & quantity required for the Development of the Software Application & Helpdesk with the manpower as mentioned in Annexure -11 of this document.

3.3 Technical Requirements of the Project

S.N.	Minimum Requirement
Platform, Hosting & Architecture	
1.	The software application will be developed on the latest stable CMS or enterprise framework (such as Drupal, Joomla, SharePoint, Liferay, Tomcat, JBoss, or WebSphere) and deployed through a complete turnkey hosting, commissioning, and O&M setup managed by the PSU.
2.	It will operate on a cloud-based, scalable architecture incorporating load balancing,

S.N.	Minimum Requirement
	failover, and redundancy, and will follow a modular, microservices-friendly design that is fully ready for containerization.
3.	The system will integrate seamlessly with external platforms through APIs and data exchange mechanisms such as XML/JSON, REST, and SOAP, while all data will be stored using secure standard formats supported by regular backup and archival processes.
Security Requirements	
4.	The system will fully comply with GoI/ GoMP security guidelines and GIGW standards and will undergo annual security audits conducted by CERT-In or STQC-approved agencies.
5.	All communication will be secured using SSL/TLS with mandatory HTTPS redirection, and the application will strictly avoid hardcoded IPs or passwords while enforcing role-based access control (RBAC) across the UI, backend, and APIs.
6.	Comprehensive audit logs will record all actions and access attempts, and the platform will remain protected through adherence to secure coding practices, timely vulnerability remediation, and prompt application of security patches.
Standards & Protocol Support	
7.	Support for major protocols: HTTP/HTTPS, SSL, XML, HTML5, REST, SOAP.
8.	Compliance with WCAG 2.0 AA, W3C, mobile governance & e-Governance standards.
9.	Support for formats: PDF, Office, XML, CSV, MHTML, images, podcasts.
10.	Interoperability with AD/LDAP for authentication.
11.	IPv6 compliance and cloud-ready design.
Performance & Optimization	
12.	The application will be fully browser and device-independent with a responsive design that works seamlessly across mobiles and tablets and will leverage asynchronous programming techniques such as lazy loading and code splitting for faster performance.
13.	It will incorporate efficient caching mechanisms at the browser, server, and distributed levels (e.g., Redis or Memcached), supported by an optimized database using indexing, query optimization, and connection pooling.
14.	Comprehensive monitoring tools will track system uptime, performance, errors, user engagement, and web performance insights, while analytics dashboards and MIS reports will provide actionable operational intelligence.
Logging, Monitoring & Analytics	
15.	Centralized logging for activities, errors, and security events.
16.	Tools for: o Search analytics o Performance monitoring
17.	MIS dashboards & exportable reports (Excel/Word/PDF).
Backup, DR & Infrastructure	
18.	Regular automated backups (data, code, documents).

S.N.	Minimum Requirement
19.	Verified restore procedures; DR tools to reduce downtime.
20.	Authentication with encryption at application & database layers.
21.	Unified client-server platform for ticketing and reservation systems.
RBAC (Role-Based Access Control)	
22.	Granular permissions (read/write/update/delete/execute).
23.	Role hierarchies; multiple roles per user.
24.	Enforcement across UI, API, backend services.
25.	Real-time permission updates; full audit trail.
26.	Automated/manual testing for access control rules.
Development, Deployment & Maintenance	
27.	The system will follow an agile delivery model with continual updates and will be built on a microservices-ready modular architecture adopting secure coding standards, with all changes formally documented through request IDs.
28.	Both code and database will be maintained under version control, supported by CI/CD pipelines enabling automated testing, safe rollbacks, and advanced deployment mechanisms such as feature flags and canary releases.
29.	A consistent UI theme will be enforced across all modules to ensure a uniform and seamless user experience.
Search Functionality	
30.	Bilingual search across the web and mobile Application.
31.	Full-text search with indexing of PDFs, MS Office files, HTML, etc.
32.	Search results filtered based on user access rights.
General Software Application Features	
33.	The system will be a CMS-based multilingual (Hindi & English) software application with full Unicode support, offering a modern, mobile-responsive UI built using HTML5, CSS, JavaScript and frameworks such as React or Angular, along with easy theme/skin switching and light/dark mode.
34.	It will support rich artwork and media, including photos, videos, and animations to provide secured login for both internal and external users with role-based access and personalized content based on user profiles.
35.	A dynamic homepage will showcase updates, events, alerts, announcements, and highlights, while the platform will maintain SEO-optimized structure with clean permalinks and a fast, filter-enabled full-text search engine.
36.	All interactive forms such as login, registration, and feedback will be protected using CAPTCHA to ensure enhanced security.
Additional Requirements for Mobile Application	
37.	The mobile app must be developed for at least two major mobile operating systems, Android (version 10 and above) and iOS (version 14 and above)
38.	Application must be capable enough to facilitate all the expected users simultaneously without any delay or limitation in the required functionalities based on the type and role of the user.
39.	The application shall be seamlessly integrated in real time with all back-end

S.N.	Minimum Requirement
	Applications as defined the Scope of Work or as per the requirement of the MPTD.
40.	Responsible to function across varying screen sizes and resolutions including smartphones, tablets etc.
41.	Mobile App should fulfill the needs of the OEMs as per the Scope mentioned in the RFP.
42.	The app shall support the required third-party integration like API, SDK etc. and comply with the applicable data protection laws and standards like WCAG etc.
43.	PSU shall regularly provide and push security and other approved updates on for the mobile application.
44.	The application shall incorporate robust security features, including secure session management and full compliance with the latest OWASP Top 10 vulnerabilities for authentication, Cert-in vulnerability, authorization and data handling.
45.	The backend system must provide role based and permission-based access to the backed application.
46.	The backed system shall provide the detailed and summary reports, dashboards etc. to MPTD as per requirement.
47.	Digital Wallet Integration: Rechargeable wallet linked to UPI, credit / debit cards, and net banking.
48.	Hosting of the mobile application on the Play Store/App Store is the responsibility of the selected bidder.
49.	The hosting cost or domain purchase cost shall be borne by the bidder, and the bidder shall also provide multiple options for the domain name to the Authority for final selection.
Development of Testing & Staging Environment	
50.	The Bidder (PSU) shall establish and maintain robust Development, Test, and Staging environments that are logically and physically independent from each other and from the Production environment. These environments must support on-demand data masking and ensure that any production data used for testing is fully masked and protected. Each component of the system shall be thoroughly tested to ensure a bug-free and stable solution.
51.	The Development and Test environments must include configuration-management capabilities and tools for version control, documentation, change management, and controlled deployment of solution components. Adequate workstations, desktops, software tools, and utilities shall be provided to support all development, testing, staging, and deployment activities.
52.	The Development environment shall be dedicated to coding activities and provide secure access for developers.
53.	The Test environment shall be used for testing software components and must be a scaled-down replica of the Production setup.
54.	The Staging environment shall support functional testing, UAT, stress testing, and performance benchmarking. Post Go-Live, the Production environment shall not be used for testing or training under any circumstances.
55.	A comprehensive test-management system must be implemented to log and track test

S.N.	Minimum Requirement
	results, maintain test cases, and record and monitor identified issues or bugs throughout the lifecycle.

Note:

Please note that as per the Expression of Interest (EOI) issued by MPTD on 08-10-2025 for the Empanelment of Manufacturers of Reflective Tapes and Rear Marking Plates, the empanelment process for the manufacturers is currently underway. Some relevant sections from the EOI are highlighted below for reference.

i) "The Empanelled OEM will intend to establish their fitment centres. These fitment centres will be equipped with power, water, IT equipment like a computer and printer, and racks for inventory management. The cost of establishing and operating the fitment centres will be borne by the OEM.

ii) The OEM will use this Management Information System (MIS) for managing inventory at the Fitment Centres and issuing QR code enabled certificates (hereinafter referred to as RRT certificate) generated from this MIS to the customers. The cost of issuing a RRT certificate / QR code will be paid by the OEM to the MIS provider. The OEM shall exclude this cost while quoting price of its products. The cost of issuing a QR certificate is subject to variation in future as approved by the State Government."

4 Roles & Responsibility

The Roles & Responsibility of the Stakeholders are mentioned below:

S.N.	Tasks	Responsibility
1	End-to-end development and deployment of MIS console, QR-Code generation, Mobile & Web based application	Selected PSU
2	Integration with Vahan Portal & Creation of API	Selected PSU
3	Development of Complaint Management System, Technical Support & Helpdesk along with all necessary equipment, furniture, peripherals, adequate and trained Manpower, setting up of required infrastructure for the helpdesk in the premises	Selected PSU
4	Deployment of Manpower as per Annexure-11	Selected PSU
5	All necessary Software, Licenses etc. necessary for smooth development and subsequent operation of entire platform	Selected PSU
6	Training to the MPTD/MPYPIL Officials as required by MPTD/MPYPIL.	Selected PSU
7	District-Wise fitment centres with Manpower, power, water, IT equipment like a computer and printer, Internet Connectivity, and racks for Inventory management etc.	Empanelled Manufacturers (OEM)
8	Cost of computer systems, Data centre infra, Servers, Switches, UTM, Routers, UPS, other hardware requirement	Selected PSU
9	UAT & Security Audit	Selected PSU